

IOQM Problem Set

Geometry

Shounak Kar

Here are some practice problems for IOQM in Geometry. The answer to each question is an integer in the range 0-99.

Problem 1

In rectangle $ABCD$, points E and F lie on sides AB and CD respectively such that both AF and CE are perpendicular to diagonal BD . Given that BF and DE separate $ABCD$ into three polygons with equal area, and that $EF = 1$, find the nearest positive integer to the length of BD .

Problem 2

Let $ABCD$ be a trapezoid with $AB \parallel CD$, $AB = 5$, $BC = 9$, $CD = 10$, and $DA = 7$. Lines BC and DA intersect at point E . Let M be the midpoint of CD , and let N be the intersection of the circumcircles of $\triangle BMC$ and $\triangle DMA$ (other than M). Compute $\lfloor EN \rfloor$.

Problem 3

Let $ABCD$ be a rectangle and E be a point on segment AD . We are given that quadrilateral $BCDE$ has an inscribed circle ω_1 that is tangent to BE at T . The incircle ω_2 of ABE is also tangent to BE at T . If the ratio of the radius of ω_1 to the radius of ω_2 is

$$\frac{a + \sqrt{b}}{c}$$

, where a, b, c are mutually co-prime positive integers, find $a + b + c$.

Problem 4

Let AEF be a triangle with $EF = 20$ and $AE = AF = 21$. Let B and D be points chosen on segments AE and AF , respectively, such that BD is parallel to EF . Point C is chosen in the interior of triangle AEF such that $ABCD$ is cyclic. If $BC = 3$ and $CD = 4$, then the ratio of areas $\frac{[ABCD]}{[AEF]}$ can be written as $\frac{a}{b}$ for relatively prime positive integers a, b . Find a .

Problem 5

In triangle ABC , let M be the midpoint of BC , H be the orthocentre, and O be the circumcentre. Let N be the reflection of M over H . Suppose that $OA = ON = 11$ and $OH = 7$. Find the remainder when BC^2 is divided by 100.

Problem 6

In triangle ABC , it is given that angles B and C are congruent. Points P and Q lie on $\overline{AC}$ and $\overline{AB}$, respectively, so that $AP = PQ = QB = BC$. Angle ACB is r times as large as angle APQ , where r is a positive real number. Find $\lfloor 100r \rfloor$.

Problem 7

The points A , B and C lie on the surface of a sphere with centre O and radius 20. It is given that $AB = 13$, $BC = 14$, $CA = 15$, and that the distance from O to $\triangle ABC$ is

$$\frac{m\sqrt{n}}{k}$$

, where m , n , and k are positive integers, m and k are relatively prime, and n is not divisible by the square of any prime. Find $m + k$.

Problem 8

In quadrilateral $ABCD$, $\angle BAD \cong \angle ADC$ and $\angle ABD \cong \angle BCD$, $AB = 8$, $BD = 10$, and $BC = 6$. The length CD may be written in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Problem 9

In triangle ABC the medians $\overline{AD}$ and $\overline{CE}$ have lengths 18 and 27, respectively, and $AB = 24$. Extend $\overline{CE}$ to intersect the circumcircle of ABC at F . The area of triangle AFB is $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Problem 10

Triangle ABC is isosceles with $AC = BC$ and $\angle ACB = 106^\circ$. Point M is in the interior of the triangle so that $\angle MAC = 7^\circ$ and $\angle MCA = 23^\circ$. Find $\angle CMB$ in degrees.

Problem 11

Let ω be a circle with centre O and radius 8, and let A be a point such that $AO = 17$. Let P and Q be points on ω such that line segments $\overline{AP}$ and $\overline{AQ}$ are tangent to ω . Let B and C be points chosen on $\overline{AP}$ and $\overline{AQ}$, respectively, such that $\overline{BC}$ is also tangent to ω . Compute the perimeter of triangle $\triangle ABC$.

Problem 12

Let circles ω_1 and ω_2 intersect at P and Q . Let the line externally tangent to both circles that is closer to Q touch ω_1 at A and ω_2 at B . Let point T lie on segment $\overline{PQ}$ such that $\angle ATB = 90^\circ$. Given that $AT = 6$, $BT = 8$, and $PT = 4$, compute $\lfloor PQ \rfloor$.

Problem 13

Circle Γ has radius 10, centre O , and diameter $\overline{AB}$. Point C lies on Γ such that $AC = 12$. Let P be the circumcentre of $\triangle AOC$. Line $\overleftrightarrow{AP}$ intersects Γ at Q , where Q is different from A . Then the value of $\frac{AP}{AQ}$ can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Compute $m + n$.

Problem 14

Let $\triangle ABC$ have $AB = 15$, $AC = 20$, and $BC = 21$. Suppose ω is a circle passing through A that is tangent to segment BC . Let point $D \neq A$ be the second intersection of AB with ω , and let point $E \neq A$ be the second intersection of AC with ω . Suppose DE is parallel to BC . If $DE = \frac{a}{b}$, where a, b are relatively prime positive integers, find $a - 10b$.

Problem 15

Hexagon $ABCDEF$ has an inscribed circle Ω that is tangent to each of its sides. If $AB = 12$, $\angle FAB = 120^\circ$, and $\angle ABC = 150^\circ$, and if the radius of Ω can be written as $m + \sqrt{n}$ for positive integers m, n , find $m + n$.

Problem 16

Let ω be a circle. Let E be on ω and S be outside ω such that line segment SE is tangent to ω . Let R be on ω . Let line SR intersect ω at B other than R , such that R is between S and B . Let I be the intersection of the bisector of $\angle ESR$ with the line tangent to ω at R ; let A be the intersection of the bisector of $\angle ESR$ with ER . If the radius of the circumcircle of $\triangle EIA$ is 10, the radius of the circumcircle of $\triangle SAB$ is 14, and $SA = 18$, then IA can be expressed in simplest form as $\frac{m}{n}$. Find $m + n$.

Problem 17

Triangle ABC has $\angle A = 90^\circ$, $AB = 2$, and $AC = 4$. Circle ω_1 has centre C and radius CA , while circle ω_2 has centre B and radius BA . The two circles intersect at point E , different from point A . Point M is on ω_2 and in the interior of ABC , such that BM is parallel to EC . Suppose EM intersects ω_1 at point K and AM intersects ω_1 at point Z . What is the area of quadrilateral $ZEBK$?

Problem 18

Triangle ABC has positive integer side lengths with $AB = AC$. Let I be the intersection of the bisectors of $\angle B$ and $\angle C$. Suppose $BI = 8$. If x the smallest possible perimeter of $\triangle ABC$, compute the sum of digits of x .

Problem 19

The circumcircle of acute $\triangle ABC$ has centre O . The line passing through point O perpendicular to $\overline{OB}$ intersects lines AB and BC at P and Q , respectively. Also $AB = 5$, $BC = 4$, $BQ = 4.5$, and $BP = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Problem 20

Triangle ABC has side lengths $AB = 7$, $BC = 8$, and $CA = 9$. Circle ω_1 passes through B and is tangent to line AC at A . Circle ω_2 passes through C and is tangent to line AB at A . Let K be the intersection of circles ω_1 and ω_2 not equal to A . Then $AK = \frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Problem 21

Let ABC be a triangle with $AB = 13$, $BC = 14$, $CA = 15$. Let H be the orthocentre of ABC . Find the distance between the circumcentres of triangles AHB and AHC .

Problem 22

Equilateral triangles $\triangle ABC$ and $\triangle DEF$ are drawn such that points B, E, F , and C lie on a line in this order, and point D lies inside triangle $\triangle ABC$. If $BE = 14$, $EF = 15$, and $FC = 16$, compute AD .

Problem 23

Let ABC be a triangle with $\angle BAC = 90^\circ$. Let D, E , and F be the feet of altitude, angle bisector, and median from A to BC , respectively. If $DE = 3$ and $EF = 5$, compute the length of BC .

Problem 24

Let Ω and ω be circles with radii 123 and 61, respectively, such that the centre of Ω lies on ω . A chord of Ω is cut by ω into three segments, whose lengths are in the ratio $1 : 2 : 3$ in that order. Given that this chord is not a diameter of Ω , compute the length of this chord.

Problem 25

Let ABC be an acute triangle. Let D, E , and F be the feet of altitudes from A, B , and C to sides BC, CA , and AB , respectively, and let Q be the foot of altitude from A to line EF . Given that $AQ = 20$, $BC = 15$, and $AD = 24$, the perimeter of triangle DEF can be written as $a\sqrt{b}$, where a, b are natural numbers and b is prime. Compute $a + b$.

Problem 26

Let $\overline{AB}$ be a diameter of circle ω . Extend $\overline{AB}$ through A to C . Point T lies on ω so that line CT is tangent to ω . Point P is the foot of the perpendicular from A to line CT . Suppose $AB = 18$, and let m denote the maximum possible length of segment BP . Find $m^2/6$.

Problem 27

In triangle ABC , $AB = AC = 100$, and $BC = 56$. Circle P has radius 16 and is tangent to $\overline{AC}$ and $\overline{BC}$. Circle Q is externally tangent to P and is tangent to $\overline{AB}$ and $\overline{BC}$. No point of circle Q lies outside of $\triangle ABC$. The radius of circle Q can be expressed in the form $m - n\sqrt{k}$, where m, n , and k are positive integers and k is the product of distinct primes. Find $m + n + k$.

Problem 28

Triangle ABC has $AC = 450$ and $BC = 300$. Points K and L are located on $\overline{AC}$ and $\overline{AB}$ respectively so that $AK = CK$, and $\overline{CL}$ is the angle bisector of angle C . Let P be the point of intersection of $\overline{BK}$ and $\overline{CL}$, and let M be the point on line BK for which K is the midpoint of $\overline{PM}$. If $AM = 180$, find LP .

Problem 29

In rectangle $ABCD$, $AB = 100$. Let E be the midpoint of $\overline{AD}$. Given that line AC and line BE are perpendicular, find the remainder when $\lfloor AD \rfloor$ is divided by 100.

Problem 30

In $\triangle ABC$ with $AB = 12$, $BC = 13$, and $AC = 15$, let M be a point on $\overline{AC}$ such that the incircles of $\triangle ABM$ and $\triangle BCM$ have equal radii. Then $\frac{AM}{CM} = \frac{p}{q}$, where p and q are relatively prime positive integers. Find $p + q$.

Problem 31

Triangle ABC with right angle at C , $\angle BAC < 45^\circ$ and $AB = 4$. Point P on $\overline{AB}$ is chosen such that $\angle APC = 2\angle ACP$ and $CP = 1$. The ratio $\frac{AP}{BP}$ can be represented in the form $p + q\sqrt{r}$, where p , q , r are positive integers and r is not divisible by the square of any prime. Find $p + q + r$.

Problem 32

In rectangle $ABCD$, $AB = 12$ and $BC = 10$. Points E and F lie inside rectangle $ABCD$ so that $BE = 9$, $DF = 8$, $\overline{BE} \parallel \overline{DF}$, $\overline{EF} \parallel \overline{AB}$, and line BE intersects segment $\overline{AD}$. The length EF can be expressed in the form $m\sqrt{n} - p$, where m , n , and p are positive integers and n is not divisible by the square of any prime. Find $m + n + p$.

Problem 33

In triangle ABC , $AB = 125$, $AC = 117$ and $BC = 120$. The angle bisector of angle A intersects $\overline{BC}$ at point L , and the angle bisector of angle B intersects $\overline{AC}$ at point K . Let M and N be the feet of the perpendiculars from C to $\overline{BK}$ and $\overline{AL}$, respectively. Find MN .

Problem 34

In triangle ABC , $AB = 20$ and $AC = 11$. The angle bisector of $\angle A$ intersects BC at point D , and point M is the midpoint of AD . Let P be the point of the intersection of AC and BM . The ratio of CP to PA can be expressed in the form $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m + n$.

Problem 35

Point P lies on the diagonal AC of square $ABCD$ with $AP > CP$. Let O_1 and O_2 be the circumcentres of triangles ABP and CDP respectively. Given that $AB = 12$ and $\angle O_1PO_2 = 120^\circ$, then $AP = \sqrt{a} + \sqrt{b}$, where a and b are positive integers. Find $a + b$.

Problem 36

Let $\triangle ABC$ be a right triangle with right angle at C . Let D and E be points on $\overline{AB}$ with D between A and E such that $\overline{CD}$ and $\overline{CE}$ trisect $\angle C$. If $\frac{DE}{BE} = \frac{8}{15}$, then $\tan B$ can be written as

$$\frac{m\sqrt{p}}{n}$$

, where m and n are relatively prime positive integers, and p is a positive integer not divisible by the square of any prime. Find $m + n + p$.

Problem 37

In $\triangle ABC$, $AC = BC$, and point D is on $\overline{BC}$ so that $CD = 3 \cdot BD$. Let E be the midpoint of $\overline{AD}$. Given that $CE = \sqrt{7}$ and $BE = 3$, the area of $\triangle ABC$ can be expressed in the form $m\sqrt{n}$, where m and n are positive integers and n is not divisible by the square of any prime. Find $m + n$.

Problem 38

In rectangle $ABCD$, $AB = 6$ and $BC = 16$. Points P, Q are chosen on the interior of side AB such that $AP = PQ = QB$, and points R, S are chosen on the interior of side CD such that $CR = RS = SD$. Find the area of the region formed by the union of parallelograms $APCR$ and $QBSD$.

Problem 39

Let $ABCDEF$ be a regular hexagon of side length 3. Let X, Y , and Z be points on segments AB, CD , and EF such that $AX = CY = EZ = 1$. The area of triangle XYZ can be expressed in the form

$$\frac{a\sqrt{b}}{c}$$

where a, b, c are positive integers such that b is not divisible by the square of any prime and $\gcd(a, c) = 1$. Find $a + b + c$.

Problem 40

Let $ABCD$ be a square with side length 16 and centre O . Let S be the semicircle with diameter AB that lies outside of $ABCD$, and let P be a point on S so that $OP = 12$. Find the remainder when the area of triangle CDP is divided by 100.

Problem 41

Let A, B, M, C, D be distinct points on a line such that $AB = BM = MC = CD = 6$. Circles ω_1 and ω_2 with centres O_1 and O_2 and radius 4 and 9 are tangent to line AD at A and D respectively such that O_1, O_2 lie on the same side of line AD . Let P be the point such that $PB \perp O_1M$ and $PC \perp O_2M$. Determine the value of $PO_2^2 - PO_1^2$.

Problem 42

Let ABC be a triangle with incentre I such that $AB = 20$ and $AC = 19$. Point $P \neq A$ lies on line AB and point $Q \neq A$ lies on line AC . Suppose that $IA = IP = IQ$ and that line PQ passes through the midpoint of side BC . Suppose that $BC = \frac{m}{n}$ for relatively prime positive integers m and n . Compute $m + n$.

Problem 43

Let ω be a circle with diameter AB and centre O . We draw a circle ω_A through O and A , and another circle ω_B through O and B ; the circles ω_A and ω_B intersect at a point C distinct from O . Assume that all three circles $\omega, \omega_A, \omega_B$ are congruent. If $CO = \sqrt{3}$, what is the perimeter of $\triangle ABC$?

Problem 44

Given a triangle ABC , consider the semicircle with diameter EF on BC tangent to AB and AC . If $BE = 1$, $EF = 24$, and $FC = 3$, find the perimeter of $\triangle ABC$.

Problem 45

In triangle ABC , $AB = 3$, $AC = 5$, and $BC = 7$. Let E be the reflection of A over BC , and let line BE meet the circumcircle of ABC again at D . Let I be the incentre of $\triangle ABD$. Given that $\cos^2 \angle AEI = \frac{m}{n}$, where m and n are relatively prime positive integers, determine $m + n$.

Problem 46

In triangle ABC , $AB = 8$, $AC = 9$, and $BC = 10$. Let M be the midpoint of BC . Circle ω_1 with area A_1 passes through A, B , and C . Circle ω_2 with area A_2 passes through A, B , and M . Then $\frac{A_1}{A_2} = \frac{m}{n}$ for relatively prime positive integers m and n . Find $m - n$.

Problem 47

A circle ω has centre O and radius r . A chord BC of ω also has length r , and the tangents to ω at B and C meet at A . Ray AO meets ω at D past O , and ray OA meets the circle centred at A with radius AB at E past A . If x is the degree measure of $\angle DBE$, find sum of digits of x .

Problem 48

In triangle ABC , $AB = 1$ and $AC = 2$. Suppose there exists a point P in the interior of triangle ABC such that $\angle PBC = 70^\circ$, and that there are points E and D on segments AB and AC , such that $\angle BPE = \angle EPA = 75^\circ$ and $\angle APD = \angle DPC = 60^\circ$. Let BD meet CE at Q , and let AQ meet BC at F . If M is the midpoint of BC , compute the degree measure of $\angle MPF$.

Problem 49

In convex quadrilateral $AEBC$, $\angle BEA = \angle CAE = 90^\circ$ and $AB = 15$, $BC = 14$ and $CA = 13$. Let D be the foot of the altitude from C to AB . If ray CD meets $\overline{AE}$ at F , compute $AE \cdot AF$.

Problem 50

Let ABC be a triangle with $AB = 5$, $AC = 8$, and $BC = 7$. Let D be on side AC such that $AD = 5$ and $CD = 3$. Let I be the incentre of triangle ABC and E be the intersection of the perpendicular bisectors of $\overline{ID}$ and $\overline{BC}$. Suppose $DE = \frac{a\sqrt{b}}{c}$ where a and c are relatively prime positive integers, and b is a positive integer not divisible by the square of any prime. Find $a + b + c$.

Bonus

In scalene $\triangle ABC$, I is the incentre, I_a is the A -excentre, D is the midpoint of arc BC of the circumcircle of ABC not containing A , and M is the midpoint of side BC . Extend ray IM past M to point P such that $IM = MP$. Let Q be the intersection of DP and MI_a , and R be the point on the line MI_a such that $AR \parallel DP$. Given that $\frac{AI_a}{AI} = 9$, the ratio $\frac{QM}{RI_a}$ can be expressed in the form $\frac{m}{n}$ for two relatively prime positive integers m, n . Compute $m + n$.

Answer Key

- | | | |
|---------------|---------------|---------------|
| 1. 2 | 18. 9 | 35. 96 |
| 2. 9 | 19. 23 | 36. 18 |
| 3. 10 | 20. 11 | 37. 10 |
| 4. 49 | 21. 14 | 38. 56 |
| 5. 88 | 22. 26 | 39. 28 |
| 6. 57 | 23. 20 | 40. 36 |
| 7. 23 | 24. 42 | 41. 65 |
| 8. 69 | 25. 19 | 42. 41 |
| 9. 63 | 26. 72 | 43. 6 |
| 10. 83 | 27. 85 | 44. 84 |
| 11. 30 | 28. 72 | 45. 55 |
| 12. 6 | 29. 41 | 46. 67 |
| 13. 89 | 30. 45 | 47. 9 |
| 14. 86 | 31. 7 | 48. 25 |
| 15. 36 | 32. 36 | 49. 99 |
| 16. 97 | 33. 56 | 50. 13 |
| 17. 20 | 34. 51 | |

Bonus **11**